When the Treatment Talks Back: Information Limits of Observational Logs from Adaptive Generative Political Communication*

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Abstract

Large language models now conduct persuasive political conversations that adapt turn by turn to the human they are persuading. We formalize the treatment delivered by such a system as a composite object — a strategy policy π, a generation kernel G, and an information environment E — and define deployment, policy, and turn-level excursion estimands for it. We then show that conversation logs collected under a deployed policy carry structurally limited information about turn-level causal effects. First, when the deployed policy has history-deterministic components, turn-level effects are nonparametrically unidentified (support failure); in RLHF-tuned deployments this condition is structurally frequent and design-reinforced. Second, even with formal positivity, we derive the efficient influence function for the turn-level excursion estimand and show that its semiparametric efficiency bound grows exponentially in a *policy decisiveness* index $D(h) = -\log \min_a \pi(a \mid h)$ — a functional of the behaviour policy alone, identified from logs, with the softmax temperature as one special case. The classical average-treatment-effect bound is recovered as a degenerate case. We then show that this is not merely a large constant: along policy sequences whose decisiveness grows fast enough relative to n, a two-point argument gives a minimax lower bound that is bounded away from zero, so no estimator recovers the turn-level effect at any rate. Third, the overlap weight mass concentrates on the policy's learned decision boundary as the policy sharpens: the surviving estimand is a boundary effect β^{ov} — formally a regression-discontinuity estimand at a threshold the policy learned rather than one the analyst chose — and we prove this in one dimension and, via a coarea argument, on a decision manifold in $\mathbb{R}^d$. We are explicit that this is localization, not stability: the surviving mass is itself $O(\tau)$. The same argument identifies the deceptively stable-but-biased regime seen in simulation as two-point indistinguishability rather than estimator pathology; where estimation becomes *easy* again is a separate achievability question we leave open. Re-querying three RLHF-tuned open-weights models on 200 real debate contexts locates plausible deployments in the difficult region, with the important caveat that the measured decisiveness is a property of a label projection and is sensitive to the judge (Section 4.4). We propose the generative micro-randomized trial (G-MRT) — within-conversation turn-level randomization with a locked evidence bank, gated generation, and assignment-ITT analysis — as the design remedy.

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1 Introduction

Conversational AI persuades. In randomized trials, GPT-4 debates shifted opponents' positions more often than human debaters did (Salvi et al., 2025), and conversations with frontier models moved policy attitudes with durable effects (Hackenburg et al., 2025). Yet measured persuasion effects are small in absolute terms: meta-analytic evidence from field experiments in political persuasion puts average effects well under one percentage point (Kalla and Broockman, 2018). Small average effects are what make decomposition — *which turn-level behaviors of the AI cause belief change* — the scientifically and politically urgent question, since an aggregate near zero is consistent with offsetting turn-level behaviors of either sign. The strongest lever in Hackenburg et al. (2025), a reward model selecting responses turn by turn, is precisely the component whose interior remains a black box: it is an adaptive turn-level policy, and its logs are what an analyst inherits.

This paper asks what those logs can and cannot identify. Our answer is pessimistic in a precise way and constructive in a precise way.

Pessimistic: the treatment in these systems is not a message but a composite (π, G, E) — a strategy policy, a generation kernel, and an information environment. Turn-level contrasts computed from deployment logs face three distinct failures: (i) support failure where the policy is history-deterministic (Proposition 1); (ii) an exponential collapse of the semiparametric efficiency bound in the policy's decisiveness even when positivity formally holds, derived from the efficient influence function of the excursion estimand itself, with a multi-turn compounding result under an additional recurrence condition and, under an additional recurrence condition, compounding across turns; and (iii) localization — the overlap mass that survives concentrates on the policy's decision boundary, so what remains estimable longest is a boundary estimand β^{ov}, distinct from the population effect and formally a regression-discontinuity quantity. We verify each mechanism in simulation, locate real deployments on the resulting (n, τ) phase diagram using re-query experiments on three RLHF-tuned models, and audit instruction–behavior slippage in 56,831 deployed conversations.

Constructive: each failure corresponds to a specific design element of a generative micro-randomized trial (G-MRT): uniform turn-level randomization restores support and overlap; exogenous assignment removes sequential confounding; regime-defined estimands resolve ill-defined treatments; and a fidelity protocol with assignment-ITT analysis handles generated-treatment noncompliance.

Contributions, three: (1) an estimand framework for generative treatments with a general adaptive action space, and the efficient influence function for its turn-level excursion estimand (Proposition 6); (2) collapse and localization indexed by a *policy decisiveness* functional that is identified from logs — an efficiency bound growing exponentially in it (Proposition 2), a matching minimax impossibility along drifting policy sequences (Proposition 4), and localization of the surviving overlap mass onto the decision boundary in one dimension (Proposition 5) and on a manifold in $\mathbb{R}^d$ (Corollary 1); (3) the G-MRT design with a generated-treatment fidelity protocol. Everything else — generator transport, internal-representation methods, simultaneous inference, constrained policy learning — is explicitly out of scope (Section 9).

2 Related Work

Texts as treatments, static and sequential. Imai and Nakamura (2026) identify effects of features of generated texts using internal representations of the generating model. Overlap is not free in that setting: it is *established*, under a separability condition on the representation, and the authors are explicit that adjusting naively on learned representations can destroy it. Nakamura and Imai (2026) extend the framework to *sequences* of treatment features — including their position within a text or video — estimating a marginal structural model with a jointly learned deconfounder for each feature in the sequence, so the difficulty that later features depend on earlier ones is squarely within their scope as well as ours. The distinction we draw is therefore not "static versus sequential" but where the loss of overlap originates. In that line the object is a generated artifact produced by a fixed procedure, and overlap is a property of the representation the analyst adjusts on. Here the object is an adaptive policy responding to a human interlocutor, and overlap is destroyed by the deployed policy's own decisiveness before the analyst does anything — which is why our results are bounds on what any analysis of such logs can achieve, and are complementary to, rather than competitive with, their estimation framework.

Off-policy evaluation with deficient support. The estimation-side face of our problem is studied in the contextual-bandit and reinforcement-learning literatures: doubly robust off-policy evaluation (Dudík et al., 2011; Thomas and Brunskill, 2016; Jiang and Li, 2016), and, closest to Proposition 1, off-policy learning under *deficient support*, where the logging policy assigns zero or near-zero probability to some actions (Sachdeva et al., 2020). We draw the boundary in four places. First, that literature works at the level of specific estimators and their variance or regret; Propositions 2 and 4 are statements about the semiparametric efficiency bound and about the minimax risk, which closes the "use a better estimator" escape by construction. Second, we index the collapse by a decisiveness functional of the policy that is identified from logs, and obtain a closed form for it in our main design. Third, localization (Proposition 5) — *where* the surviving information lives, and the $\tau \downarrow 0$ limit of the overlap-weighted estimand — has, to our knowledge, no counterpart in that literature, whose remedies (truncation, regularized weights, support restriction) implicitly change the estimand without characterizing the limit object. Fourth, our remedy is a *design* (randomize the turns), not an estimator: the setting is scientific inference about turn-level persuasion effects, where the analyst can run an experiment, not a fixed logged dataset to be salvaged.

Micro-randomized trials. MRTs randomize pre-defined discrete actions at decision points and target causal excursion effects via WCLS (Qian et al., 2022; Boruvka et al., 2018). We import the excursion estimand and add what conversation requires: a generation kernel between assignment and realization, ill-defined observational labels, and fidelity auditing of generated treatments. We make no claim to be first to combine language models with micro-randomization; our novelty claim is confined to within-conversation turn-level randomization of generative political persuasion with information locking.

Stochastic interventions. Regime-specific potential outcomes for stochastic and modified policies (Díaz Muñoz and van der Laan, 2012; Kennedy, 2019; Young et al., 2014) supply our formal object: β is defined under a kernel intervention — assign the label, realize from G. We derive the efficient influence function for the factorial excursion contrast in that framework (Proposition 6) rather than importing a bound stated for the average treatment effect, and our substantive addition is the decisiveness collapse that follows from it.

Overlap weighting. β^{ov} connects to overlap weights and empirical equipoise (Li et al., 2018); Proposition 5 and Corollary 1 give the sharp-policy limit of that estimand, which is a boundary effect in the regression-discontinuity sense (Hahn et al., 2001; Keele and Titiunik, 2015) at a threshold the policy learned rather than one the analyst chose.

Information equivalence. Dafoe et al. (2018) are a caution rather than a licence: they argue that manipulations of one attribute routinely carry information about others, and are pessimistic about isolating a focal dimension. We take the warning literally — it is why we do not call dose "information" but a high-information communication package, and why the fidelity audit measures what the instruction actually changed rather than assuming it changed only one thing.

Support failure is old; its systematicity is new. Nonidentification from deterministic treatment assignment is standard since Robins (1986). Our claim is not the structure but its origin: RLHF preference optimization widens score gaps and deployment favors low temperature, so history-determinism in deployed LLM policies is *structurally frequent and design-reinforced* — it cannot be escaped by collecting better logs from the same deployment.

3 Generative Treatments

Individuals $i = 1, \ldots, n$ converse over turns $t = 1, \ldots, T$. At turn t the system draws an action from its policy, $A_{it} \sim \pi_t(\cdot \mid H_{it})$, where H_{it} is the interaction history, and realizes an utterance from its generation kernel, $M_{it} \sim G_\theta(\cdot \mid A_{it}, H_{it}, E_j)$, where E_j is the information environment (for us: a locked evidence bank on issue j). The user responds R_{it}; outcomes Y_i include belief accuracy, calibration, reactance, and one-week retention.

The action space is general. $A_t \in \mathcal{A}$ may be a discrete strategy label, a continuous generation parameter, or a factorial cell. As a running example — and as the implementation object of the companion experimental paper — we use a 2×2: form $F \in \{\text{assertive}, \text{interrogative}\} \times$ dose $D \in \{\text{high: 4–5 facts}, \text{low: 1–2 facts}\}$.

Treatment is a stochastic intervention. Potential outcomes are defined under regimes, $Y_i^{\pi,G}$, with do-notation read as a kernel intervention: intervene on the label, realize from G (Díaz Muñoz and van der Laan, 2012; Young et al., 2014). Three estimand layers:

- **Deployment effect** $\Delta^{\text{dep}} = \mathbb{E}[Y^{\pi_1,G_1}] - \mathbb{E}[Y^{\pi_0,G_0}]$: what existing RCTs identify.
- **Policy effect** $\Delta^{\text{pol}}(G) = \mathbb{E}[Y^{\pi_1,G}] - \mathbb{E}[Y^{\pi_0,G}]$: holding the generator fixed.
- **Turn-level excursion effects** (primary): contrasts of one turn's action against a pre-specified reference policy ρ (uniform over $\mathcal{A}$, all t, all h), with subsequent turns following ρ. Proximal excursions (immediate reactance, belief update) are standard CEE/WCLS targets (Boruvka et al., 2018); distal excursions (final accuracy, retention) connect to distal causal excursion effect estimation (Qian, 2025) — using ordinary WCLS for terminal outcomes is a known attack point we avoid by construction.

Ill-defined treatment in logs. An observational label $\tilde{S}_t = s$ corresponds to the realized kernel $M_t \sim G(\cdot \mid s, H_t)$, a *different distribution at every history.* Label-pooled contrasts do not correspond to any well-defined treatment pair. Our regime definition does not "solve" this nonidentification; it replaces the question with a well-posed one — and the well-posed β is identified only under assignment randomization (Section 5).

4 Why Transcripts Are Not Enough

Model the deployed policy's action choice as softmax in a history-dependent score gap with effective temperature τ: for the two-action case, $p_\tau(h) = \sigma(\Delta(h)/\tau)$ with σ the logistic function. Deployment practice pushes τ down; preference optimization pushes $|\Delta|$ up.

4.1 Support failure

Proposition 1 (Support failure)**.** *If $\pi_t^{\text{dep}}(a \mid h) = 0$ on a positive-probability set of histories, turn-level excursion effects are nonparametrically unidentified.*

The counterexample pair requires no latent confounder — the failure is support, not confounding. Take $T = 1$, $H \in \{h_0, h_1\}$, $\pi^{\text{dep}}(1 \mid h_0) = 1/2$, $\pi^{\text{dep}}(1 \mid h_1) = 0$, $Y(a) \mid H = h \sim N(\mu_a(h), 1)$, and let two DGPs differ only in $\mu_1(h_1) = c_1$ versus c_2. The observed-data law factorizes as $P(H)\, \pi^{\text{dep}}(A \mid H)\, N(y; \mu_A(H), 1)$, to which the $(1, h_1)$ stratum contributes nothing; the two DGPs are observationally identical yet their uniform-reference excursion effects differ by $P(h_1)(c_2 - c_1)$. The multi-turn version replaces H with $(X, R_{1:t-1})$ and recovers the time-varying-confounding form. The structure is Robins (1986); the LLM-specific claim is frequency and reinforcement, not inevitability.

4.2 Exponential collapse of the information bound

With $\tau > 0$, positivity holds formally, so the right response to Proposition 1 alone would be "use IPW." Proposition 2 closes that escape at the level of the efficiency bound.

Proposition 2 (Result A)**.** *Let $D(h) = -\log \min_{a \in \mathcal{A}^*} \pi(a \mid h)$ be the policy's decisiveness at h (Definition 1), where $\mathcal{A}^*$ is the set of actions entering the contrast. Assume (A1$'$) $\mathbb{P}\{D(H) \geq d\} \geq c > 0$; (A2$'$) conditional outcome variances on that set are bounded below by $\underline{\sigma}^2$; and (A3) the model class is nonparametric in the outcome regression — no parametric extrapolation into unobserved strata. Then the semiparametric efficiency bound for the turn-level excursion effect satisfies*

$$V \;\geq\; \underline{\sigma}^2\, \rho_{\min}^2\, \mathbb{E}\big[e^{D(H)}\big] \;\geq\; \underline{\sigma}^2\, \rho_{\min}^2\, c\, e^d,$$

with $\rho_{\min}$ the smallest reference probability on $\mathcal{A}^*$.

We derive this from the efficient influence function of the excursion estimand itself (Proposition 6 and Appendix A) rather than importing a bound stated for the average treatment effect. The ATE bound of Hahn (1998) is recovered as the two-action, degenerate-reference special case. Two consequences are worth isolating. First, the governing quantity is a property of the behaviour policy alone and is identified from logs; a softmax at temperature τ gives $D = |\Delta|/\tau$ as one special case, but the statement covers truncated and constrained decoding, hard filters, and learned $\arg\max$ rules, none of which are softmax. Second, in a factorial design the relevant probabilities are the *cell* probabilities, not the marginal for the factor being contrasted: because the reference regime draws the other factor from ρ, a design in which the form-marginal stays at $1/2$ while cells become rare still has an exploding bound.

By the convolution theorem, *no regular estimator in the class* attains lower asymptotic variance: switching from IPW to AIPW recovers no counterfactual information that the sample does not contain. Parametric extrapolation escapes the bound, but that is buying identification with an assumption, and should be said in exactly those words. Effective sample size is arm-specific Kish: $n_{\text{eff}}(a) = n/\mathbb{E}[1/\pi_\tau(a \mid H)]$. The slogan "halving τ squares the required n" holds within fixed-gap strata; with heterogeneous gaps the max-gap stratum dominates the rate.

Remark 1 (Single-turn bounds are a fortiori multi-turn bounds). Our results are stated for one turn and two actions, while the motivating setting is a T-turn conversation. The gap runs in the harmless direction. The turn-t excursion estimand conditions on the observed history H_t and contrasts actions at that turn, with subsequent turns following ρ. The observed-data model factorizes turn by turn; the turn-t factor is exactly the single-turn problem with propensity $\pi_\tau(\cdot \mid H_t)$, and identifying β_t additionally requires bridging every later turn from π^{dep} to ρ. Discarding the information cost of those later-turn bridges can only *understate* the difficulty; hence any lower bound for the single-turn problem at the (Δ, τ) prevailing at turn t is also a lower bound for the conversational problem. (K-action policies reduce to the two-action case via the two arms of the contrast.)

Proposition 3 (A-T: multi-turn compounding)**.** *Consider the T-turn policy value $V(\rho)$ of the uniform reference ρ over K actions, observed under π_τ. Start from the semiparametric efficiency bound for finite-horizon off-policy evaluation with history-dependent policies (Kallus and Uehara, 2020; Jiang and Li, 2016):*

$$V_{\text{eff}} = \mathrm{Var}(V_1(H_1)) + \sum_{t=1}^{T} \mathbb{E}_\pi\Big[\Big(\prod\nolimits_{s\le t} w_s\Big)^2 \mathrm{Var}(V_{t+1} \mid H_t, A_t)\Big], \qquad w_s = \tfrac{\rho(A_s|H_s)}{\pi_\tau(A_s|H_s)}.$$

Assume (M1) uniform recurrence of the high-gap set*: $\mathbb{P}(H_{s+1} \in \mathcal{H}_\delta \mid h, a) \ge p_\delta > 0$ for every $h \in \mathcal{H}_\delta$ and every action a; and (M2) a per-turn conditional variance floor $\underline{\sigma}^2$ on $\mathcal{H}_\delta$. Then*

$$V_{\text{eff}} \;\ge\; \underline{\sigma}^2 \cdot \mathbb{P}(H_1 \in \mathcal{H}_\delta) \cdot p_\delta^{T-1} \cdot \left[\frac{e^{\delta/\tau}}{K^2}\right]^T,$$

which compounds geometrically in T whenever $\tau < \delta/\log(K^2/p_\delta)$.

Proof sketch. All terms of V_{eff} are nonnegative; keep the terminal one. On $\mathcal{H}_\delta$, $\mathbb{E}_\pi[w_t^2 \mid H_t = h] = K^{-2}\sum_a 1/\pi_\tau(a \mid h) \ge K^{-2}e^{\delta/\tau}$ since the rarest action has probability at most $e^{-\delta/\tau}$. Induct backward from $t = T$: by the tower property and (M1) — whose action-uniformity is what lets the bound pass through each turn's weight — each turn contributes a factor $p_\delta\, e^{\delta/\tau}/K^2$, and turn 1 contributes $\mathbb{P}(H_1 \in \mathcal{H}_\delta)\, e^{\delta/\tau}/K^2$. We quote rather than re-derive the finite-horizon efficiency bound; the form above is the discrete-horizon Cramér–Rao bound of Jiang and Li (2016), which Kallus and Uehara (2020) extend to history-dependent policies. Our contribution is the insertion and the induction. □

Remark 2 (Scope of Proposition 3). Two limitations should be read with this proposition. It bounds the *policy value* $V(\rho)$, which we treat as a secondary estimand (Section 3); the corresponding statement for the primary turn-level excursion effect requires redoing the induction with the continuation weights of Appendix A and we do not claim it here. And (M1) requires the history to remain in $\mathcal{H}_\delta$ under *every* action, i.e. that the treatment does not move the history state much — close to the opposite of the premise in this paper's title. The per-turn factor is $p_\delta e^{\delta/\tau}/K^2$, and p_δ is not estimated anywhere in this paper, so multi-turn compounding should be read as conditional on an unverified quantity rather than as an empirical claim.

Substantively, (M1) says decisiveness persists regardless of what the system does — as with strongly reactant partisan users. The K^2 factor reflects the conservativeness of the uniform reference and is not optimized; the excursion version (turn t, reference thereafter) follows with exponent $T-t+1$. The honest reading: more turns make the information limit exponentially worse, now as a proposition rather than a heuristic.

Proposition 4 (Minimax collapse)**.** *Let $M(\pi) := \sup_{d>0} \mathbb{P}\{D(H) \geq d\}\, e^d$. Consider the model class in which π is known and fixed, $Y \mid A = a, H = h \sim N(\mu_a(h), \sigma^2)$, and $|\mu| \leq B$. Then for any estimator $\hat{\beta}$,*

$$\inf_{\hat{\beta}} \sup_{P} \; \mathbb{E}_P\big|\hat{\beta} - \beta(P)\big| \;\geq\; \tfrac{1}{4} \min\Big\{ B\rho_{\min},\; \rho_{\min}\, \sigma \sqrt{M(\pi)/n} \,\Big\}.$$

In particular, along a sequence of policies with $M(\pi_n)/n \to \infty$ the minimax risk is bounded away from zero: no estimator recovers β at any rate.

This is the statement that Proposition 2 alone does not support. Proposition 2 bounds the information constant at a fixed policy, where β remains regularly $\sqrt{n}$-estimable; Proposition 4 says that once decisiveness grows fast enough relative to sample size, estimability itself fails. The proof (Appendix A) is a two-point argument in which the two laws differ only in the outcome regression at the rarest action on the high-decisiveness stratum, so that they are separated in the estimand but nearly indistinguishable in the data. For the main design $H \sim U(-1,1)$, $M = \tau e^{(1-\tau)/\tau}$ in closed form, i.e. $\log M = 1/\tau - 1 - \log(1/\tau)$.

The deceptive regime is this indistinguishability. In the main simulation design, two laws whose true ATEs differ by 0.198 — differing only in β on the rarely-treated region — are separated by 0.16 standard deviations of the Hájek estimator at $\tau = 0.08$, $n =$ 2,000, and still only 0.40 at $n =$ 32,000, a sixteenfold increase in sample size. At $\tau = 0.22$ the same 0.200 gap is recovered as 0.224 at 1.69 standard deviations. What the phase diagram labels "biased but stable" is not a description of estimator behaviour but an instance of two-point indistinguishability.

4.3 Localization, the counterexample, and the phase diagram

Proposition 5 (Result B, one-dimensional)**.** *Let $\omega_\tau(H) = p_\tau(1-p_\tau)$ and $\beta_\tau^{\mathrm{ov}} = \mathbb{E}[\omega_\tau\{Y(1) - Y(0)\}]/\mathbb{E}[\omega_\tau]$. Assume:*

- *(B1) H has a density f, continuous in a neighbourhood of each root;*
- *(B2) $\Delta \in C^1$ and the root set $\mathcal{R} = \{h : \Delta(h) = 0\}$ is* finite*, with every root simple ($\Delta'(r) \neq 0$);*
- *(B3) $c(h) = \mathbb{E}[Y(1) - Y(0) \mid H = h]$ is bounded, and continuous near each root;*
- *(B4) $f(r) > 0$ for at least one $r \in \mathcal{R}$.*

together with the tail-separation condition (B2′): $\inf_{\mathrm{dist}(h,\mathcal{R}) \geq \epsilon} |\Delta(h)| > 0$ for every $\epsilon > 0$. Then, as $\tau \downarrow 0$, the normalized overlap measure $\omega_\tau f / \int \omega_\tau f$ converges weakly to $\sum_j w_j \delta_{r_j}$ with weights $w_j \propto f(r_j)/|\Delta'(r_j)|$. In the special case of a unique *root at 0 the limit is a single point mass and*

$$\mathbb{E}[\omega_\tau] = \tau \, \frac{f(0)}{|\Delta'(0)|} \, (1 + o(1)), \qquad \beta_\tau^{\mathrm{ov}} \to \mathbb{E}[Y(1) - Y(0) \mid H = 0].$$

Corollary 1 (Decision-manifold localization, $H \in \mathbb{R}^d$)**.** *Let $H \in \mathbb{R}^d$ with density f, let $\mathcal{B} = \{h : \Delta(h) = 0\}$, and assume: (C1) $\Delta \in C^1$ and f is bounded and continuous near $\mathcal{B}$; (C2) 0 is a* regular value *of Δ, i.e. $\nabla\Delta(h) \neq 0$ on $\mathcal{B}$ — the d-dimensional analogue of the simple-root condition (B2); (C3)–(C4) the tail-separation condition of Proposition 5 holds with* $\mathrm{dist}(\cdot, \mathcal{B})$*; and (C5) $0 < \int_{\mathcal{B}} f/\|\nabla\Delta\| \, d\mathcal{H}^{d-1} < \infty$. Then as $\tau \downarrow 0$,*

$$\mathbb{E}[\omega_\tau] \;=\; \tau \int_{\mathcal{B}} \frac{f}{\|\nabla\Delta\|} \, d\mathcal{H}^{d-1} \, (1 + o(1)),$$

and the normalized overlap measure converges weakly to the measure on $\mathcal{B}$ with $\mathcal{H}^{d-1}$-density proportional to $f(h)/\|\nabla\Delta(h)\|$. Consequently $\beta^{\mathrm{ov}}_\tau \to \int_{\mathcal{B}} c\, f/\|\nabla\Delta\| \big/ \int_{\mathcal{B}} f/\|\nabla\Delta\|$. Proposition 5 is the case $d = 1$, where $\mathcal{H}^0$ is counting measure and $\|\nabla\Delta\| = |\Delta'|$.

Remark 3 (What β^{ov} is)*.* The limit in Corollary 1 is a boundary estimand in the regression-discontinuity sense: in one dimension it is exactly $\mathbb{E}[Y(1) - Y(0) \mid \Delta(H) = 0]$, the conditional effect at the assignment threshold (Hahn et al., 2001), and in d dimensions it is the $f/\|\nabla\Delta\|$-weighted average effect over a boundary, the object studied in multi-dimensional and geographic RD (Keele and Titiunik, 2015). The substantive reading is sharper than an overlap-weighting analogy: logs from a sharply adaptive system behave less like an observational study than like a natural experiment at a threshold — but the threshold is one the *policy* learned, not one the analyst chose. What is stably estimable therefore describes the histories on which the policy is *undecided*, which in a decisive deployment is precisely the minority of traffic.

Remark 4 (The surviving information also vanishes)*.* Proposition 5 says *where* overlap mass concentrates, not how much of it there is. Since $\mathbb{E}[\omega_\tau] = O(\tau)$, the effective sample available for β^{ov}_τ is itself $O(n\tau)$ and degenerates in the same limit. Result B should therefore not be read as saying that boundary information is *stable*: what survives longest is boundary information, but it too disappears as the policy sharpens. Assumption (B4) is exactly what prevents the leading term from collapsing to $0 = o(\tau)$; without it the statement is vacuous.

Proof idea. $\omega_\tau = K(\Delta(h)/\tau)$ exactly, with $K(u) = 1/[4\cosh^2(u/2)]$ and $\int K = 1$: an approximate-identity kernel. The tail-separation condition kills contributions away from the roots at rate $e^{-c(\epsilon)/\tau}$; near a simple root, the change of variables $v = \Delta(h)$, $v = \tau u$ and dominated convergence give $\int g\, \omega_\tau f \, dh = \tau\, g(0) f(0)/|\Delta'(0)| + o(\tau)$ for bounded continuous g. Ratios deliver the three claims. With multiple roots, mass distributes proportionally to $f(r_j)/|\Delta'(r_j)|$. Full proof in Appendix C; the multivariate coarea version is Corollary 1, proved in Appendix D. □

Why the boundary estimand matters substantively. β^{ov} is not a consolation prize; it is the effect in a policy-relevant population. The decision boundary $\{\Delta(H) = 0\}$ collects the histories at which the deployed policy is genuinely undecided — the algorithmic analogue of clinical equipoise (Li et al., 2018). Three consequences. First, *marginal policy evaluation*: any small change to the policy — a regulation, a guardrail, a reward-model update, a temperature change — alters behavior precisely on and near this boundary, so the effect among boundary histories is the first-order term in the value change of any local policy modification. Second, *auditability*: β^{ov} is the one turn-level causal quantity a regulator could estimate from logs alone with stable precision, which makes its scope and its limits worth stating exactly. Third, the political reading, stated as interpretation rather than result: histories far from the boundary — strongly reactant partisan contexts where the policy is most decisive — are exactly where population-level causal information vanishes; what deployment data can tell us about persuasion is confined to the persuadable margin as *the policy* defines it, not as the scientist would.

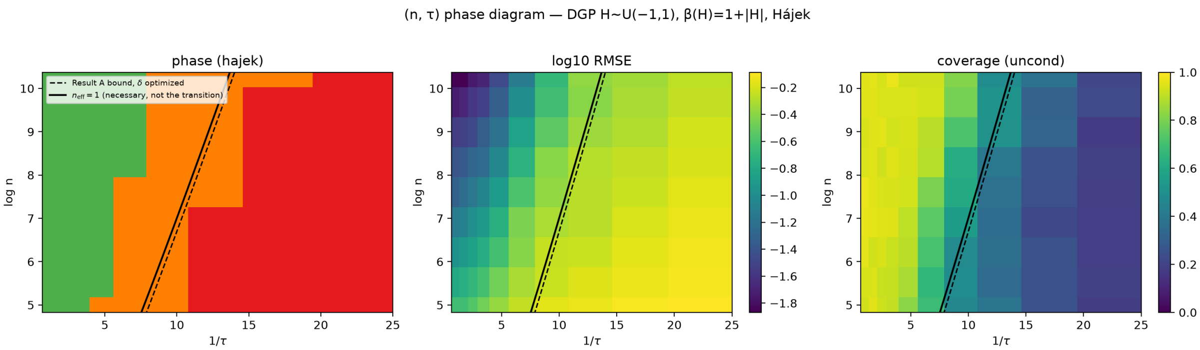

Figure 1: (n, τ) phase diagram; DGP $H \sim U(-1,1)$, $\beta(H) = 1 + |H|$, Hájek estimator, 400 replications per cell. Left: regime classification (green estimable, orange variance explosion, red biased-stable). Middle, right: $\log_{10}$ RMSE and unconditional coverage. Two reference lines, neither of which is the observed transition. Dashed: the Proposition 2 bound with δ optimized, $\log n = 1/\tau - 1 - \log(1/\tau)$. Solid: the $n_{\text{eff}} = 1$ contour $\log \mathbb{E}[1/p_\tau] = \log(1 + \tau \sinh(1/\tau))$, a necessary condition only; measured onsets of estimability lie 3–5 nats above it. Regime thresholds (relative bias 0.10/0.20, coverage 0.85) are presentation choices defended by the RMSE and coverage panels.

What Result B does *not* say. At fixed $\tau > 0$, an IPW estimator with true propensities targets the *population* effect as $n \to \infty$ — the estimand does not silently become the boundary effect because the policy is sharp. Numerically: $\tau = 0.3$, $n = 4 \times 10^6$, $\beta(H) = 1 + |H|$ gives IPW $1.792 \approx$ ATE $1.798 \neq \beta^{\text{ov}} = 1.356$. Formal identification and practical estimability are different properties; our claims are about the *rate* of information loss and the *location* of surviving information, never about a change of target.

The joint limit is the real object. The failure surface lives in (n, τ): with $n \gg e^{\delta/\tau}$ the population effect is estimable (at exploding variance); near $n \sim e^{\delta/\tau}$ variance explodes; with $n \ll e^{\delta/\tau}$ rare crossover actions never appear in the sample and estimators become *biased while looking stable* — the deceptive regime. Figure 1 shows the three regimes for the main design $H \sim U(-1,1)$, $\Delta(H) = H$, with two *reference* lines. The dashed line is the Proposition 2 bound with δ chosen optimally: since $\mathbb{P}(\mathcal{H}_\delta) = 1 - \delta$ here, $\max_\delta (1-\delta)e^{\delta/\tau}$ is attained at $\delta = 1 - \tau$, giving $\log n = 1/\tau - 1 - \log(1/\tau)$. (Fixing $\delta = 1$, as an earlier version did, puts $\mathbb{P}(\mathcal{H}_\delta) = 0$ and makes the bound identically zero.) The solid line is the $n_{\text{eff}} = 1$ contour $\log \mathbb{E}[1/p_\tau] = \log(1 + \tau \sinh(1/\tau))$, exact for this design. We stress that neither line is the observed transition: the $n_{\text{eff}} = 1$ contour is a *necessary* condition, and on a 13×13 grid the measured onsets of estimability lie 3–5 nats above it. Which functional of the decisiveness distribution governs the transition is open (Section 9).

4.4 Where real deployments sit: Study 0 (descriptive, motivating)

Three empirical bridges locate deployed systems on the diagram. Their epistemic status is *motivating*: they estimate proxies of policy decisiveness and instruction–behavior slippage, not the parameters of Sections 4.1–4.3.

(0-a) Reproduction. Our pipeline reproduces the direction of Salvi et al. (2025) on the released data ($n = 750$): personalized Human-AI has the largest crude odds ratio for increased

model	median entropy (bits)	zero-entropy contexts	modal share
Qwen2.5-7B-Instruct	0.0	89.5%	REBT 98.6%
Mistral-7B-Instruct-v0.3	0.0	58.0%	REBT 94.5%
Phi-3.5-mini-instruct	0.0	75.5%	REBT 97.3%

Table 1: Re-query decisiveness across model families: 200 contexts $\times$ $k = 20$ regenerations each; strategies classified by a single 32B judge. Low entropy recurs across families — the observable signature of "structurally frequent."

agreement ($2.18 > 1$), matching the original ordering.

(0-b) Re-query decisiveness. For 200 rebuttal contexts stratified by treatment and prior-agreement strength, we regenerate $k = 20$ responses per context and classify the persuasion strategy of each. Across three RLHF-tuned open-weights families the per-context strategy distribution is near-degenerate (Table 1).

Substituting per-strategy $\hat{p}_s$ into the effective-sample formula (Dirichlet-smoothed, $k = 20$ resolution): the modal strategy needs $\sim 3 \times 10^3$ conversations per contrast arm and minority strategies are on the order of 10^5 (7.7×10^4 to 1.2×10^5 across strategies and judges), with a two-minority contrast needing the *sum* of two such terms. Three caveats are essential. First, these are illustrative insertions into Proposition 2's formula, not estimates of any deployment's policy. Second, the magnitude is set by choices we make, not by the data: $n_{\text{req}} = \sigma^2\, \mathbb{E}[1/\hat{p}]/\text{SE}^{*2}$ with $\sigma^2 = 1$ and $\text{SE}^* = 0.02$, and for a strategy never observed in k draws the smoothed $\hat{p}$ is exactly $\alpha/(k + \alpha K)$, so $\mathbb{E}[1/\hat{p}]$ is the constant $(k + \alpha K)/\alpha = 48$ regardless of how rare the strategy truly is; halving α to 0.1 moves the figure to $\sim 5 \times 10^5$. Third, for the same reason these numbers say "below the resolution of $k = 20$," not "this rare."

Placing contexts on the phase diagram. Only the composite Δ/τ — the score gap in temperature units, exactly the phase diagram's abscissa — is identified from choice frequencies; τ alone is not. Estimating the two-action reduction $\widehat{\Delta/\tau} = \log[(c_1 + \alpha)/(c_2 + \alpha)]$ per context yields a median of 3.71 for all three models — which is the *resolution ceiling* of $k = 20$ draws: 58–89.5% of contexts are right-censored there (the runner-up action was never realized). The reflexive reading is that measuring a deployed policy's decisiveness runs into the same information limit that blocks effect estimation: with the runner-up never realized, k draws cannot distinguish "rare" from "absent." We are deliberately careful about what that licenses. The value $3.71 = \log\{(k + \alpha)/\alpha\}$ is the arithmetic ceiling of the estimator at $k = 20$, not a property of any model, which is why all three models return it. A defensible one-sided statement uses the exact binomial instead: observing the runner-up 0 times in 20 draws gives a Clopper–Pearson 95% upper bound $p_2 \leq 0.139$, hence $\Delta/\tau \geq 1.82$ and $\mathbb{E}[1/p] \geq 7.2$ — an order of magnitude below the ceiling value. We therefore do not propagate the ceiling figure through Proposition 3; with $K = 8$ strategy categories that proposition's compounding condition $\Delta/\tau > \log(K^2/p_\delta)$ is not met by the measured values in any case.

Two robustness results discipline these numbers. *Judge sensitivity*: relabeling all 4,000 Qwen responses with a 32B judge yields raw agreement 0.903 but $\kappa = 0.13$, and $\kappa \approx 0$ once both-modal pairs are excluded — judges agree on the dominant label and disagree almost completely on minority-label identity. Aggregate quantities are nevertheless stable (per-strategy n_{req} ratios 0.88–1.52 across judges). We therefore report entropy, modal share, and n_{req} — and we do not interpret minority-label identity before human double-coding (pre-registered $\kappa \geq 0.6$ on 200 turns). *Label-error simulation*: propagating misclassification rates $q \in \{0.05, 0.1, 0.2\}$ through the pipeline (symmetric and majority-absorbing noise models anchored to the observed confusion pattern) moves rare-strategy n_{req} by -25% to $+6\%$ even at $q = 0.2$; symmetric noise biases toward optimism, so the 10^5 scale is conservative.

(0-c) Instruction $\neq$ behavior. In the released data of Hackenburg et al. (2025) (56,831 persuasion conversations across three studies), the randomized rhetoric instruction explains $\eta^2 = 0.20/0.12/0.32$ of realized fact-count variance in studies 1/2/3 — 68–88% of realized-dose variation occurs *within* assignment. Assignments also bundle: "information" realizes 8.3–21.8 facts on average while "moral reframing" realizes 1.6–2.2, and 52–56% of moral-reframing conversations realize zero facts. Where compliance flags exist, valence compliance is 72.7% (study 1) and 89.6% (study 3); grammar and on-topic compliance are above 92% in both. What was randomized was an instruction; the treatment that reached the participant was a generated behavior, substantially decoupled from it. This is the observational counterpart of the fidelity problem the G-MRT protocol addresses by design.

5 The G-MRT Design as the Remedy

Each failure in Section 4 maps to a design element:

failure	design element
support failure + bound collapse	uniform turn-level randomization $\rho(a \mid h) = 1/4 \geq \epsilon$
sequential confounding	exogenous assignment of A_t
ill-defined treatment	regime-defined (kernel-intervention) estimands
generated-treatment noncompliance	assignment-ITT + fidelity protocol

The protocol: (1) conversation-level randomization of the generator $Z_i \in \{G_a, G_b\}$, at least one open-weights; (2) turn-level uniform randomization of A_t with full logging of assignment, probability, and realization; (3) a locked evidence bank of ≥ 20 verified fact units per issue with per-turn sub-banks or without-replacement sampling *specified as part of the treatment regime* — dose is honestly named a high-information communication package (fact count covaries with length and cognitive load; length-controlled designs are a follow-up); (4) a fidelity gate that inspects each utterance before sending, regenerates once on mismatch, and sends-with-log on second failure. The gate is part of the treatment kernel: ITT identifies the effect of *assignment to the generate-and-gate protocol* $\tilde{G}$, and the paper says so in those words. Differential noncompliance across cells (e.g., interrogative form failing more often) contaminates cell contrasts; we pre-register a fidelity floor of $\geq 80\%$ per cell with demotion of affected contrasts, publish gate-decision logs, and pre-specify a no-gate sensitivity analysis, since a gate classifier whose errors correlate with assignment would itself distort the treatment.

The price of identification (illustrative, not a theorem). Mixing exploration into deployment, $\rho = (1-\epsilon)\pi_\tau + \epsilon\,\text{Unif}$, caps the variance contribution at $\sigma^2 K/(n\epsilon)$ at a per-turn regret cost scaling with ϵ — a design corollary quantifying the trade-off, presented as an illustration because its exact form depends on outcome bounds, estimator, and horizon.

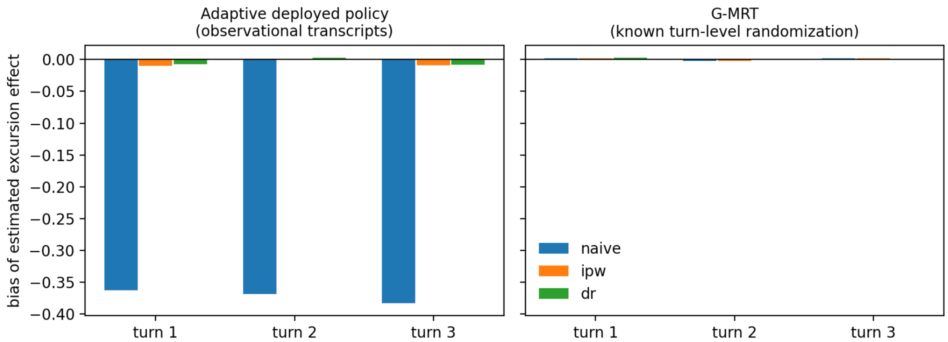

Figure 2: Sign reversal under an adaptive policy: naive turn-level contrasts versus known-ρ IPW/DR.

6 Estimation

Proximal excursions: WCLS with known randomization ρ (Boruvka et al., 2018; Qian et al., 2022). Distal excursions: DCEE estimators (Qian, 2025), not terminal-outcome WCLS. Policy values: cross-fitted doubly robust estimators for 2–3 pre-specified policy pairs only. Scope sentence: all guarantees are relative to the G-MRT's known ρ; nothing in this section rescues transcript-only analysis. Multiplicity: corrections for a pre-specified family, finalized after pilot; simultaneous confidence regions are companion-paper material.

7 Simulation Evidence

Four executed studies; theory-to-DGP correspondence is declared object-by-object in the repository.

(i) Sign reversal under adaptivity. A defensiveness-adaptive policy makes the naive turn-level contrast wrong in *sign* (truth $+0.068$; naive -0.294, coverage 0.0) while known-ρ IPW/DR recover it (bias ≤ 0.01, coverage ≈ 0.95–0.97) — the confounding face of the problem, and the G-MRT's licence (Figure 2).

(ii) Two-state temperature collapse. Along a τ grid calibrated so $\tau = 1$ reproduces the adaptive policy of (i): IPW SE grows exponentially, tracking a scaled inverse-propensity diagnostic curve (labeled as such — it is not the efficiency bound); the naive estimator becomes *more confident while more wrong* as τ falls (coverage $\to 0$ with shrinking SE) (Figure 3). Metric discipline: global arm-specific Kish ESS (theory object), minimum stratum-arm cell occupancy (rare-crossover diagnostic — not Kish), $\mathbb{E}[p(1-p)]$ (Proposition 5 object), and $\mathbb{E}[2\min(p, 1-p)]$ (diagnostic) are reported as four separate columns, coverage both conditional and unconditional on estimability, with zero-cell rates alongside. One finding we flag for practice: in the collapse regime the *realized* control-arm Kish ESS is non-monotone in τ: it falls to 0.028 at $\tau = 0.40$, while the zero-stratum-arm rate is still 0.00, and then *rebounds* to 0.037, 0.372, 0.534 at $\tau = 0.30, 0.22, 0.15$ — over exactly the range where the zero-stratum-arm rate climbs $0.06 \to 0.70 \to 0.99$. Surviving weights homogenize as rare crossovers vanish from the sample, so a healthy-looking realized ESS

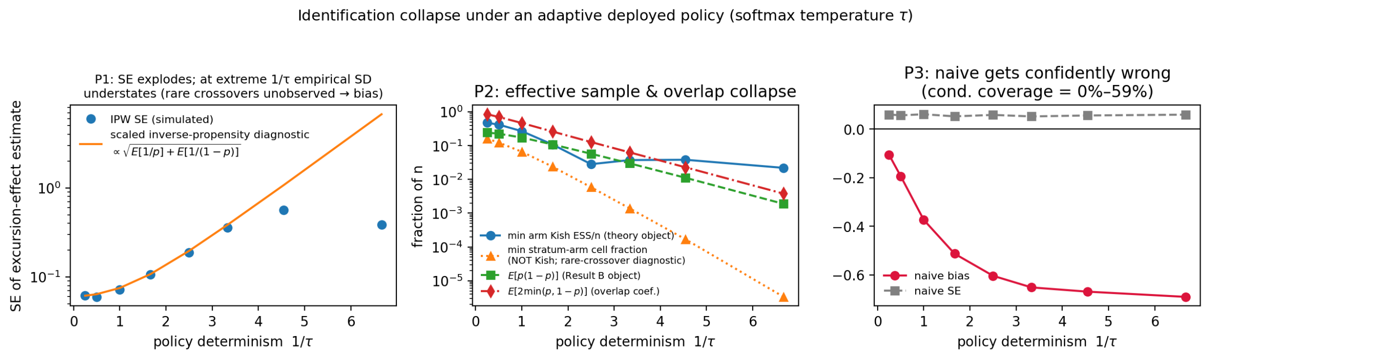

Figure 3: Two-state temperature collapse: predictions P1–P3.

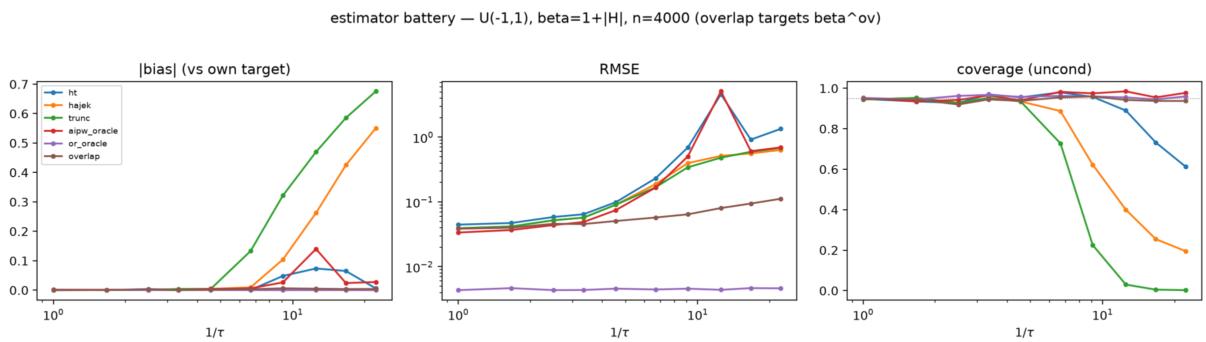

Figure 4: Estimator battery on the continuous-boundary design, $n = 4{,}000$ (bias against each estimator's own target; the overlap estimator targets β^{ov}). The numerical values quoted in the text are for $n = 32{,}000$, the largest cell of the grid.

co-occurring with high zero-cell rates is itself the signature of the deceptive regime. ESS should never be reported without zero-cell rates.

(iii) Continuous-boundary battery. ($H \sim U(-1,1)$ main, $N(0,1)$ robustness; $\beta(H) \in \{1,\, 1 + e^{-H^2},\, 1 + |H|\}$; 400 replications per cell.) At $n = 32{,}000$: all population-targeting estimators are unbiased with nominal coverage down to $\tau \approx 0.15$; at $\tau = 0.045$ Horvitz–Thompson has SD 2.95 (variance explosion) while Hájek shows bias 0.46 with SD 0.30 — biased-yet-stable; oracle AIPW retains nominal coverage throughout but does not rescue *precision*, its SD inflating from 0.012 to 1.38 (a factor of 116) as τ falls — which is the behaviour Proposition 2 predicts, and confirms that the efficiency loss is not an IPW artifact; the oracle outcome-regression line is flat by construction — the (A3) escape made visible, identification bought by assumption; and the overlap estimator remains unbiased with nominal coverage *for its own target* β^{ov} at every τ, with $\beta^{\text{ov}}_\tau \to \beta(0) = 1$ as Proposition 5 predicts (Figure 4). Under $N(0,1)$ the collapse is earlier and, at extreme τ, propensities underflow to machine 0/1 so IPW becomes numerically undefined — reported as estimability 0, not dropped.

(iv) Phase diagram. The three regimes separate cleanly over the (n, τ) grid (Figure 1). We report the reference lines without claiming either is the transition: measured onsets of estimability lie 3–5 nats above the $n_{\text{eff}} = 1$ contour, and on a widened 13×13 grid neither a mean functional

($\log \mathbb{E}[e^D]$) nor an upper-tail functional of D reproduces a stable offset across the $U(-1,1)$ and $N(0,1)$ designs. We therefore present the phase structure as a simulation finding and leave its asymptotic characterization open.

8 Empirical Plan and Ethics

The companion experimental paper implements the 2×2 G-MRT: pilot $n = 80$–120 (function verification: assignment mechanics, fidelity rates, bank leakage, dropout — not effect detection), demonstration $n = 180$–250 with pre-registered minority contrasts, confirmatory sample from post-pilot power simulation. Pre-registration locks the fidelity floor, blocking factors (issue), attrition handling (IPW-for-attrition), and reactivity checks (randomizing measurement itself in pilot). Ethics: persuasion toward *accuracy* on factual political questions with a fact-checked locked bank (cf. Costello et al., 2024); no deception arms; disclosure of AI interlocutor; IRB before pilot. The three Study 0 analyses involve only public data and no new human subjects.

9 Discussion

What we rule in: estimand definitions that survive the generative setting; explicit information limits with rates and locations; a design that restores identification at a quantified exploration cost. What we rule out of this paper: generator transport (effects across model versions — motivated empirically by cross-model heterogeneity in Chen et al., 2026), internal-representation adjustment (overlap problems of representation conditioning are themselves design-based cautionary tales, cf. Tierney et al., 2025), simultaneous inference over excursion families, and epistemically constrained policy learning. Each is a companion, not a section.

Limitations. Five, stated plainly.

The impossibility is asymptotic in the policy, not the sample. Proposition 2 bounds the information constant at a fixed policy, where the excursion effect remains regularly $\sqrt{n}$-estimable; Proposition 4 obtains a genuine impossibility only along sequences whose decisiveness grows with n. Whether a particular deployment is in that regime is an empirical question our Study 0 can motivate but not settle.

We bound impossibility, not achievability. Proposition 4 says when estimation must fail; it does not say when it becomes easy, and the observed onset of estimability in simulation is the latter. The two are separated by a design-dependent amount: the gap between the lower-bound functional M and the oracle-IPW variance factor $\mathbb{E}[e^D]$ is 1.00 nats for $U(-1,1)$ and 2.06 for $N(0,1)$. No single functional can therefore track both curves, and a matching upper bound is left open.

Proposition 3 rests on a recurrence condition in tension with our own framing. Condition (M1) requires histories to remain in the high-gap set under *every* action, i.e. that treatment does not move the history state much — which is close to the opposite of the premise in this paper's title. Its per-turn factor is $\rho^2_{\min} e^D p_\delta$, and p_δ is not estimated anywhere here; multi-turn compounding should be read as conditional on an unverified quantity.

Study 0 measures a projection, not a policy. The entropy and decisiveness figures in Section 4.4 are properties of an eight-category strategy label assigned by an LLM judge, not of the generation policy itself. Three specific threats: the modal category (rebuttal) is near-tautological with the generation prompt, which asks for a rebuttal; the two judges we ran agree at $\kappa = 0.13$

overall and not at all once both-modal pairs are excluded, and the headline entropy figures move materially between them; and at $k = 20$ regenerations a strategy with true probability 0.02 is unobserved with probability 0.67, so "never observed" is not "very rare." At the level of raw text the generations are not degenerate at all — within-context duplicates are absent and pairwise lexical overlap is low. Any collapse we report is a collapse *after* projection to the label space. The re-queried models are also open-weights stand-ins for the deployed system, not the deployed system. We therefore use Study 0 to motivate and locate, not to establish; human double-coding and higher-k re-query on the censored subset are the obvious next steps and are not yet done.

Study 0-c measures a bundled dose. Realized fact counts covary with AI verbosity ($r = 0.49$ and 0.71 in studies 2 and 3), so the within-assignment variance we report is not purely instruction–behavior slippage; adjusting for message length *raises* η^2 rather than lowering it.

Data and code availability. All numbers reported here are produced by code in the project repository, which contains the simulation sources, the analysis pipeline, the re-query outputs and judge labels, and the derived result files. The DebateGPT reproduction uses the public release of Salvi et al. (2025); the deployment audit uses the public release of Hackenburg et al. (2025), and the 56,831 conversations are the released persuasion rows, not that paper's full sample. Simulation entry points and expected runtimes are documented in the repository README.

The one-sentence version of this paper: *adaptive systems generate the least population-level causal information exactly where they are most decisive, and the information that survives lives on their decision boundary* — so if you want turn-level causal knowledge from conversational AI, you must randomize the turns.

A Efficient influence function and the decisiveness bound

Setup. Observe $O = (H, A, Y)$ with $A \in \mathcal{A}$, $|\mathcal{A}| = K < \infty$, and $\pi(a \mid h) = \mathbb{P}(A = a \mid H = h) > 0$ a.s. Write $\mu(a, h) = \mathbb{E}[Y \mid A = a, H = h]$ and $\sigma^2(a, h) = \mathrm{Var}(Y \mid A = a, H = h)$; the model is nonparametric. In the multi-turn setting read $H = H_t$, Y as the proximal outcome, and all statements on the availability stratum $\{I_t = 1\}$.

A regime is a known action distribution $q(\cdot \mid h)$ on $\mathcal{A}$ — a stochastic intervention — with value $\psi(q) = \mathbb{E}\big[\sum_a q(a \mid H)\, \mu(a, H)\big]$. The form estimand of Section 3 is $\beta^{\text{form}} = \psi(q_1) - \psi(q_0)$ with $q_1(f, d \mid h) = \mathbf{1}\{f = \text{assertive}\}\rho_D(d)$ and q_0 likewise for interrogative. Note that the dose is marginalized under the *reference* ρ_D, not under π; this is what makes the cell propensities, rather than the form-marginal, the relevant quantities.

Proposition 6 (Efficient influence function)**.** *$\psi(q)$ is pathwise differentiable with efficient influence function*

$$\varphi_q(O) \;=\; \sum_a q(a \mid H)\mu(a, H) \;-\; \psi(q) \;+\; \frac{q(A \mid H)}{\pi(A \mid H)}\big(Y - \mu(A, H)\big),$$

so that for $\beta = \psi(q_1) - \psi(q_0)$ the semiparametric efficiency bound is

$$V(\beta) \;=\; \mathrm{Var}\big(m_1(H) - m_0(H)\big) \;+\; \mathbb{E}\Big[\textstyle\sum_a \big(q_1(a \mid H) - q_0(a \mid H)\big)^2 \frac{\sigma^2(a,H)}{\pi(a|H)}\Big], \tag{1}$$

where $m_j(h) = \sum_a q_j(a \mid h)\mu(a, h)$.

Proof. $\psi(q)$ depends on the observed-data law only through the marginal of H and the outcome regression μ. Decompose the tangent space as $\mathcal{T} = \mathcal{T}_H \oplus \mathcal{T}_{A|H} \oplus \mathcal{T}_{Y|A,H}$ and differentiate along a regular parametric submodel. The $\mathcal{T}_H$ component contributes $\sum_a q\mu - \psi$; the $\mathcal{T}_{Y|A,H}$ component contributes $(q/\pi)(Y-\mu)$. Because q is a known function and ψ does not depend on π, the $\mathcal{T}_{A|H}$ component vanishes — this holds even though q may depend on h. One checks $\mathbb{E}[\varphi_q] = 0$ and $\varphi_q \in \mathcal{T}$, so φ_q is the EIF. The two components are orthogonal, giving $\mathrm{Var}(\varphi_q) = \mathrm{Var}(m_q(H)) + \mathbb{E}[(q/\pi)^2\sigma^2]$; expanding the second term over a and using linearity in q for the contrast yields (1). $\square$

When q_1 and q_0 have disjoint support — as they do for a form or dose contrast — $(q_1 - q_0)^2 = q_1^2 + q_0^2$ and the sum in (1) runs over all K cells.

Definition 1 (Policy decisiveness)**.** For $\mathcal{A}^* = \mathrm{supp}(q_1) \cup \mathrm{supp}(q_0)$ set $D(h) := -\log \min_{a \in \mathcal{A}^*} \pi(a \mid h)$.

D is a functional of the behaviour policy alone and is identified from logs; neither τ nor a softmax parameterization enters.

Proof of Proposition 2. Discard the first term of (1). On $\{D \geq d\}$, $\sum_a (q_1 - q_0)^2\sigma^2/\pi \geq \underline{\sigma}^2\rho_{\min}^2 \sum_{a\in\mathcal{A}^*} 1/\pi(a \mid H) \geq \underline{\sigma}^2\rho_{\min}^2/\min_{a\in\mathcal{A}^*}\pi(a \mid H) = \underline{\sigma}^2\rho_{\min}^2 e^{D(H)} \geq \underline{\sigma}^2\rho_{\min}^2 e^d$; multiply by $\mathbb{P}\{D \geq d\} \geq c$. $\square$

Remark 5 (Recovering Hahn (1998)). With $K = 2$, $q_1 = \delta_1$, $q_0 = \delta_0$ (so $\rho_{\min} = 1$), (1) is $\mathrm{Var}(\mu_1 - \mu_0) + \mathbb{E}[\sigma_1^2/\pi(1 \mid H) + \sigma_0^2/\pi(0 \mid H)]$, i.e. the ATE bound of Hahn (1998). The present paper therefore does not import that bound for a different estimand; it derives the bound for the excursion estimand and recovers Hahn's as a special case. If π is a softmax at temperature τ and $K = 2$, then $D = |\Delta|/\tau + O(e^{-|\Delta|/\tau})$ and the $e^{\delta/\tau}$ form is recovered; the converse fails, so Proposition 2 applies to sharply adaptive policies that are not softmax (truncated or constrained decoding, learned $\arg\max$ rules, hard filters). For $K = 2$ one also has the exact identity $1/\{p(1-p)\} = 2 + 2\cosh(\operatorname{logit} p)$.

Remark 6 (Scope of Proposition 2). Proposition 2 is a statement about the asymptotic variance at a *fixed* policy. Whenever $\pi(a \mid h) > 0$ everywhere, $V(\beta) < \infty$ and β remains regularly $\sqrt{n}$-estimable: on its own this is a constant factor, not an impossibility. The impossibility statement requires drifting sequences π_n and is Proposition 4 below; the closest antecedent for the irregular behaviour it describes is Khan and Tamer (2010).

Proof of Proposition 4. Fix $d > 0$, let $\mathcal{H}_d = \{D \geq d\}$ and $c = \mathbb{P}(\mathcal{H}_d) > 0$, and let a^* attain the minimum in Definition 1, so $\pi(a^* \mid h) \leq e^{-d}$ on $\mathcal{H}_d$. Define two laws P_0, P_1 that agree in the marginal of H, in π, and in $\mu(a,h)$ for every (a,h) except one: $\mu^{(0)}(a^*, h) = 0$ and $\mu^{(1)}(a^*, h) = \varepsilon$ for $h \in \mathcal{H}_d$.

Separation in the estimand. Since $\psi(q) = \mathbb{E}[\sum_a q(a \mid H)\mu(a,H)]$,

$$\beta(P_1) - \beta(P_0) \;=\; \mathbb{E}\big[q(a^* \mid H)\,\varepsilon\,\mathbf{1}\{H \in \mathcal{H}_d\}\big] \;\geq\; \rho_{\min}\,c\,\varepsilon \;=:\; \Delta.$$

Proximity in the data. The observed density factorizes as $P(H)\,\pi(A \mid H)\,N(Y; \mu_A(H), \sigma^2)$, and P_0, P_1 share the first two factors, so only the $(a^*, \mathcal{H}_d)$ cell contributes to the divergence:

$$\mathrm{KL}(P_0 \| P_1) = \mathbb{E}_H\Big[\textstyle\sum_a \pi(a \mid H)\,\mathrm{KL}\big(N(\mu_a^{(0)}, \sigma^2) \,\|\, N(\mu_a^{(1)}, \sigma^2)\big)\Big] = c\,\pi(a^* \mid h)\,\frac{\varepsilon^2}{2\sigma^2} \leq \frac{c\,e^{-d}\varepsilon^2}{2\sigma^2}.$$

By tensorization $\mathrm{KL}(P_0^n \| P_1^n) = n\,\mathrm{KL}(P_0 \| P_1)$. Choose $\varepsilon = \sigma\sqrt{e^d/(nc)}$, so that this equals $1/2$; Pinsker's inequality then gives $\mathrm{TV}(P_0^n, P_1^n) \le \sqrt{\mathrm{KL}/2} = 1/2$.

Le Cam. The two-point bound $\inf_{\hat\beta}\sup_P \mathbb{E}|\hat\beta - \beta| \ge (\Delta/2)(1-\mathrm{TV}) \ge \Delta/4$ with $\Delta = \rho_{\min} c\varepsilon = \rho_{\min}\sigma\sqrt{c\,e^d/n}$. Optimizing over d replaces $c\,e^d$ by $M(\pi)$. The constraint $|\mu| \le B$ caps ε at B, giving the stated minimum (for the two-point method see Tsybakov, 2009, §2.2). □ □

B Support failure (Proposition 1)

Take $T = 1$, $H \in \{h_0, h_1\}$ with $\mathbb{P}(H = h_1) > 0$, $\pi^{\mathrm{dep}}(1 \mid h_0) = 1/2$ and $\pi^{\mathrm{dep}}(1 \mid h_1) = 0$, and $Y(a) \mid H = h \sim N(\mu_a(h), 1)$. Let two laws P_1, P_2 agree in every component except $\mu_1(h_1)$, equal to c_1 under P_1 and $c_2 \ne c_1$ under P_2. The observed-data density factorizes as $P(H)\,\pi^{\mathrm{dep}}(A \mid H)\,N(y; \mu_A(H), 1)$ and the stratum $\{A = 1, H = h_1\}$ has probability zero, so $\mu_1(h_1)$ does not appear: P_1 and P_2 induce identical observed-data laws. Their uniform-reference excursion effects differ by $\mathbb{P}(h_1)(c_2 - c_1) \ne 0$. Hence no functional of the observed-data law equals the excursion effect, which is the claim. No latent confounder is used: the failure is support, not confounding. The multi-turn version replaces H by $(X, R_{1:t-1})$ and recovers the standard time-varying-confounding form of Robins (1986); the contribution here is not the structure but the claim that the triggering condition is structurally frequent in RLHF-tuned deployment, which is an empirical claim (Section 4.4) and not a theorem.

C Localization in one dimension (Proposition 5)

Throughout $\omega_\tau(h) = p_\tau(1-p_\tau) = \frac{1}{4}\mathrm{sech}^2\big(\frac{\Delta(h)}{2\tau}\big) = \frac{1}{\tau}K\big(\frac{\Delta(h)}{\tau}\big)$ with $K(u) = 1/[4\cosh^2(u/2)]$, an exact identity. Two elementary facts: $\int_{\mathbb{R}} K = [\frac{1}{2}\tanh(u/2)]_{-\infty}^{\infty} = 1$, and $\cosh^2(u/2) \ge e^{|u|}/4$ gives $K(u) \le e^{-|u|}$.

Fix a bounded continuous g and put $I_\tau(g) = \int \omega_\tau g f$. Split at $U_\epsilon = \{\mathrm{dist}(h, \mathcal{R}) < \epsilon\}$. Outside U_ϵ, (B2′) gives $|\Delta| \ge \eta(\epsilon) > 0$, so $\omega_\tau \le e^{-\eta/\tau}$ and, f and g being bounded, that contribution is $O(e^{-\eta/\tau}) = o(\tau)$. Inside, work near a single root r with $\Delta'(r) \ne 0$ (B2); by the inverse function theorem Δ is a C^1 diffeomorphism of a neighbourhood of r onto a neighbourhood of 0, so substituting $v = \Delta(h)$ and then $v = \tau u$,

$$\int_{U_\epsilon} \omega_\tau g f\,dh = \int K(u)\,\frac{(gf)(\Delta^{-1}(\tau u))}{|\Delta'(\Delta^{-1}(\tau u))|}\,\tau\,du.$$

The integrand is dominated by $K(u)\sup_{|h-r|\le\epsilon}|gf/\Delta'|$, finite because $|\Delta'|$ is continuous and bounded away from 0 near r; by dominated convergence and continuity of f, g at r the integral tends to $g(r)f(r)/|\Delta'(r)|$. Hence $I_\tau(g) = \tau\,g(r)f(r)/|\Delta'(r)| + o(\tau)$. Summing over the finitely many roots (B2) gives $I_\tau(g) = \tau\sum_j g(r_j)f(r_j)/|\Delta'(r_j)| + o(\tau)$. Taking $g \equiv 1$ yields the statement for $\mathbb{E}[\omega_\tau]$, where (B4) $f(r) > 0$ for some root is exactly what prevents the leading term from vanishing and the expansion from degenerating to $0 = o(\tau)$. Ratios $I_\tau(g)/I_\tau(1)$ give weak convergence to $\sum_j w_j\delta_{r_j}$ with $w_j \propto f(r_j)/|\Delta'(r_j)|$, and $g = c$ (bounded, continuous near the roots, by (B3)) gives the limit of β_τ^{ov}. With a unique root at 0 this is a point mass and $\beta_\tau^{\mathrm{ov}} \to c(0)$.
□

D Localization on a decision manifold (Corollary 1)

Let $H \in \mathbb{R}^d$ and $\mathcal{B} = \Delta^{-1}(0)$. By (C2), $\nabla\Delta \neq 0$ on $\mathcal{B}$, so 0 is a regular value and $\mathcal{B}$ is a C^1 hypersurface of dimension $d-1$. On $U_{\epsilon_0} = \{|\Delta| < \epsilon_0\}$ the coarea formula (e.g. Evans and Gariepy, Thm. 3.4.2) gives, for integrable Φ,

$$\int_{U_{\epsilon_0}} \Phi \, dh = \int_{\mathbb{R}} \Big[\int_{\Delta^{-1}(s)} \frac{\Phi}{\|\nabla\Delta\|} \, d\mathcal{H}^{d-1} \Big] ds.$$

Apply this with $\Phi = \omega_\tau g f$. Since ω_τ depends on h only through $\Delta(h)$, it is constant on level sets and factors out: with $G(s) := \int_{\Delta^{-1}(s)} gf/\|\nabla\Delta\| \, d\mathcal{H}^{d-1}$,

$$\int_{U_{\epsilon_0}} \omega_\tau g f \, dh = \int_{-\epsilon_0}^{\epsilon_0} \tfrac{1}{\tau} K(s/\tau) \, G(s) \, ds = \int_{|u| \le \epsilon_0/\tau} K(u) \, G(\tau u) \, du \cdot \tau.$$

By (C1)–(C2) and the implicit function theorem the level sets $\Delta^{-1}(s)$ vary continuously in s near 0 along the normal flow of $\mathcal{B}$, and f, g, $\|\nabla\Delta\|$ are continuous with $\|\nabla\Delta\|$ bounded away from 0; hence G is continuous at 0 with $G(0) = \int_{\mathcal{B}} gf/\|\nabla\Delta\| \, d\mathcal{H}^{d-1}$. Since $\int K = 1$ and $K(u) \le e^{-|u|}$, dominated convergence gives $\int_{U_{\epsilon_0}} \omega_\tau g f = \tau \, G(0) \, (1 + o(1))$. The region outside U_{ϵ_0} contributes $O(e^{-\eta/\tau})$ by (C4) as in Appendix C. Taking $g \equiv 1$ gives the stated expansion of $\mathbb{E}[\omega_\tau]$; (C5) guarantees the limit is in $(0, \infty)$, so ratios converge and the normalized overlap measure tends weakly to the measure on $\mathcal{B}$ with $\mathcal{H}^{d-1}$-density proportional to $f/\|\nabla\Delta\|$. Taking $g = c$ gives the limit of β_τ^{ov}. For $d = 1$, $\mathcal{H}^0$ is counting measure and $\|\nabla\Delta\| = |\Delta'|$, recovering Appendix C. $\square$

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